

Faculty of Computer Technology

MCA

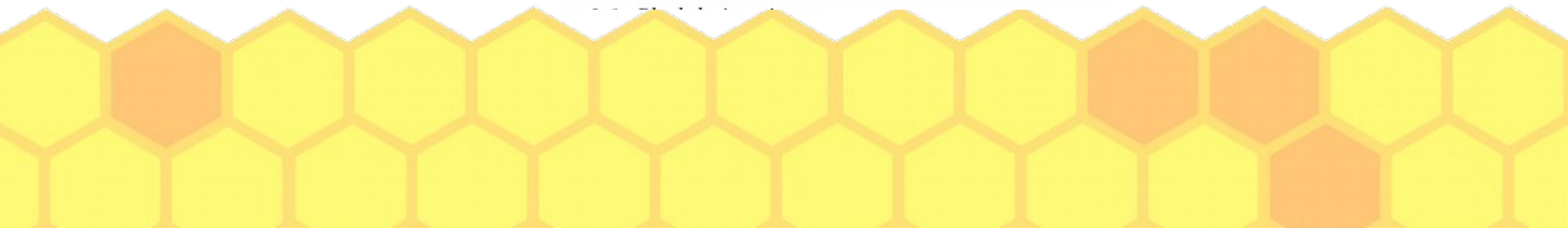
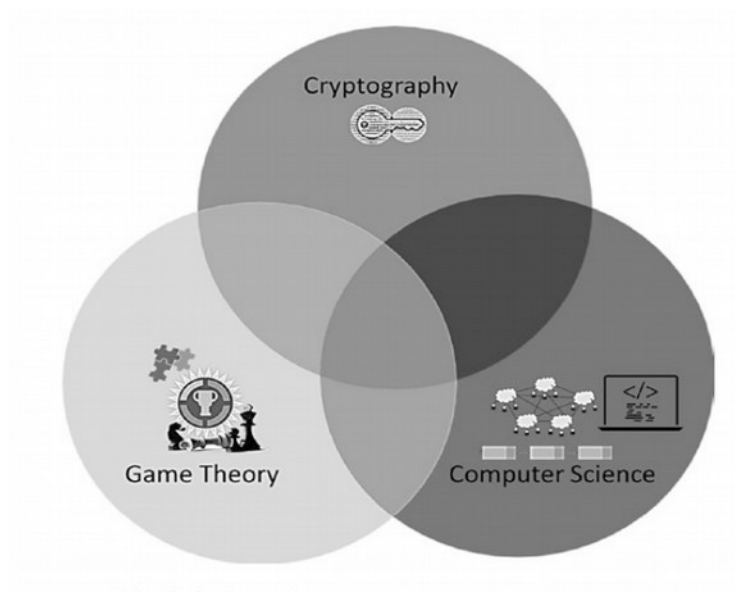
Sem – II

Block Chain Technology

Chapter-2

Laying the Blockchain Foundation

- Block chain is not just a technology, it is coupled with business functions and use cases.
- It is a brilliant amalgamation of the concepts from cryptography, game theory and computer science engineering.



Cryptography

- Usages of Cryptography:

- 1) Confidentiality
- 2) Data Integrity
- 3) Authentication
- 4) Non-repudiation



Cryptography

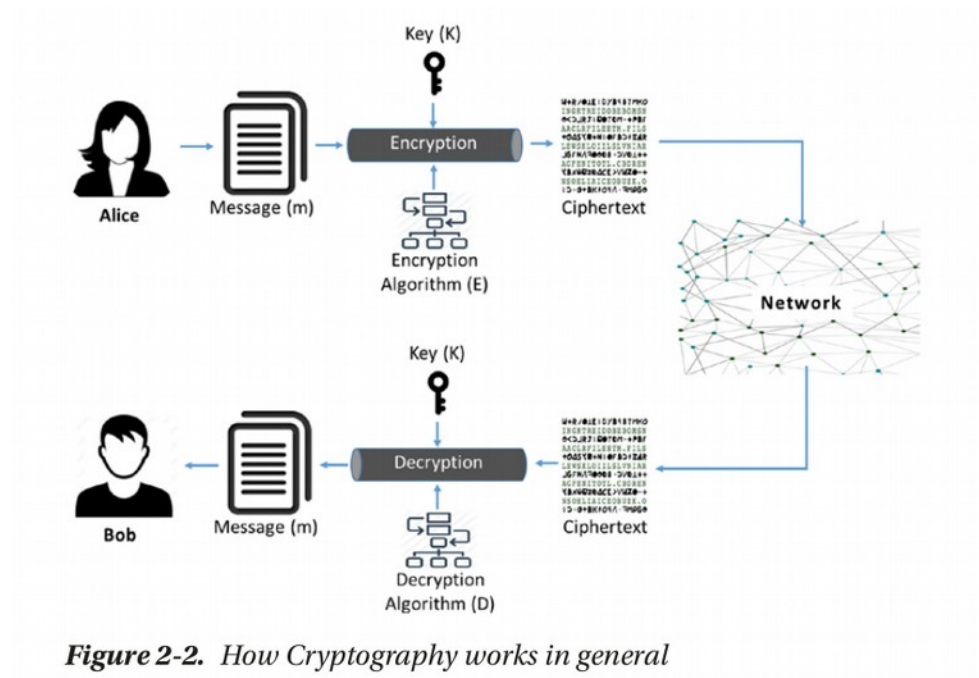


Figure 2-2. How Cryptography works in general

Types of Cryptography

- Two kinds of Cryptography:

- 1) Symmetric Key Cryptography

- 2) Asymmetric Key Cryptography (a.k.a public key)



Symmetric Key Cryptography

- If the same key is used for both encryption and decryption , it is called symmetric key cryptography.
- Sender and receiver both agrees on a key called “shared secret” before they exchange the ciphertext.

Alice—the Sender:

- Encrypt the plaintext message m using encryption algorithm E and key k to prepare the ciphertext c
- $c = E(k, m)$
- Send the ciphertext c to Bob

Bob—the Receiver:

- Decrypt the ciphertext c using decryption algorithm D and the same key k to get the plaintext m
- $m = D(k, c)$



Symmetric Key Cryptography

- Sender and receiver uses same key (k) which they share it with each other using secure communication channel.

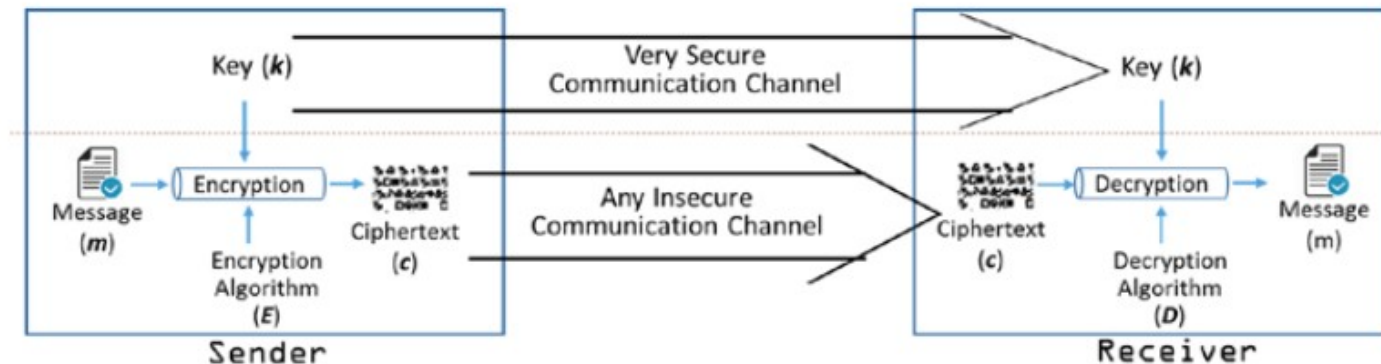


Figure 2-3. Symmetric cryptography

Symmetric Key Cryptography

- It is used widely, the most common uses are secure file transfer protocol such as HTTPS, SFTP and WebDAVS (Web Distributed Authoring and Versioning is an extension of the Hypertext Transfer Protocol (HTTP)).
- This cryptography system are usually faster and more useful when the data size is huge.
- It exists in two variants:
 - 1) Stream ciphers and
 - 2) Block Cipher



Asymmetric Key Cryptography

- It is also known as “Public Key Cryptography”.
- Introduced by Diffie and Hellman.
- It solves the problem of symmetric key cryptography’s problem of key distribution by introducing digital signature.



Asymmetric Key Cryptography



Alice—The Sender:

- Encrypt the plaintext message **m** using encryption algorithm **E** and the public key **Puk_{Bob}** to prepare the ciphertext **c**.
- $c = E(\text{Puk}_{\text{Bob}}, m)$
- Send the ciphertext **c** to Bob.

Bob—The Receiver:

- Decrypt the ciphertext **c** using decryption algorithm **D** and its private key **Prk_{Bob}** to get the original plaintext **m**.
- $m = D(\text{Prk}_{\text{Bob}}, c)$



Asymmetric Key Cryptography

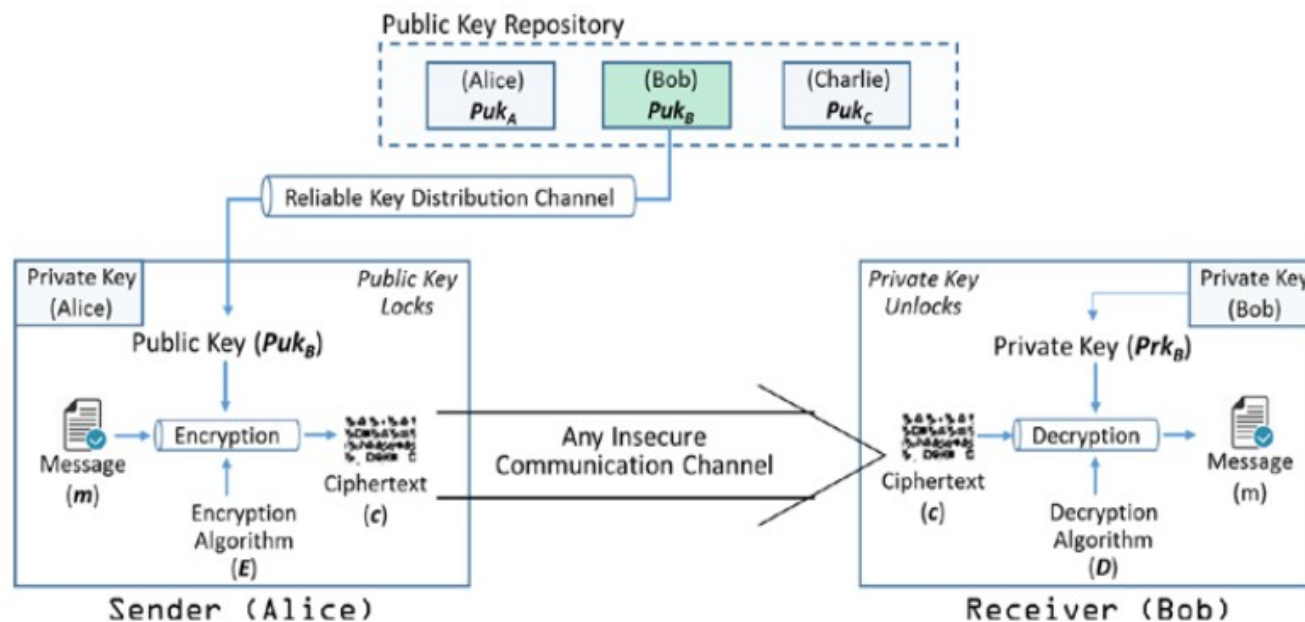


Figure 2-13. Asymmetric cryptography for confidentiality

Asymmetric Key Cryptography

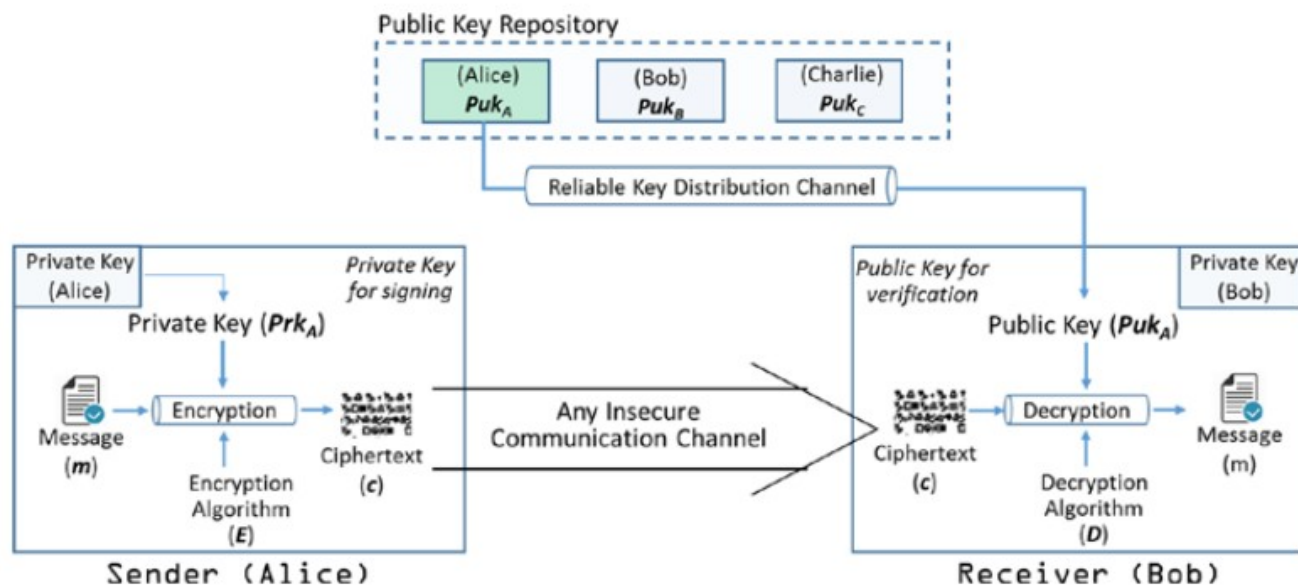


Figure 2-14. Asymmetric cryptography for authentication

ASymmetric Key Cryptography

- To facilitate confidentiality and authentication, public key encryption has to be used twice:
 - 1) The message should be first encrypted with the sender's private key to provide a digital signature.
 - 2) Then it should be encrypted with the receiver's public key to provide confidentiality.

- $c = E[\text{Puk}_{\text{Bob}}, E(\text{Prk}_{\text{Alice}}, m)]$

- $m = D[\text{Puk}_{\text{Alice}}, D(\text{Prk}_{\text{Bob}}, c)]$

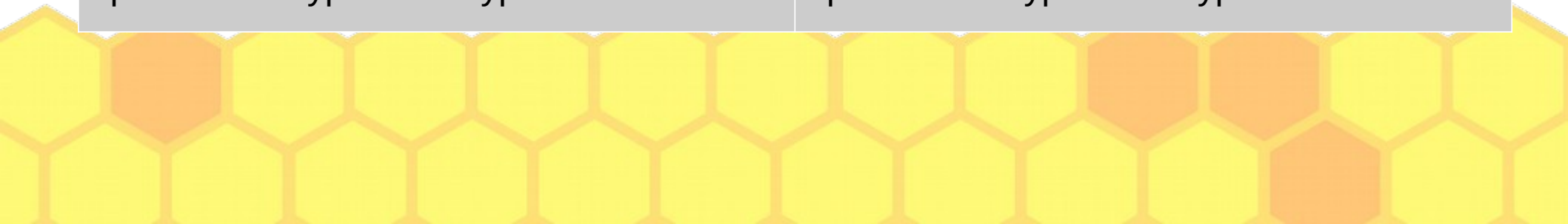


Asymmetric Key Cryptography

- The two uses of the two keys in asymmetric cryptography are:
 - 1) Public keys are known and accessible to everyone. They can be used to encrypt the message or to verify the signatures.
 - 2) Private keys are extremely private to individuals. They are used to decrypt the message or to create signatures.
- Third party Public Key Infrastructure (PKI) is introduced to provide verified public key and are assured by the process of attestation or notarization of user identity.
- Public key encryption is also known as one-way function or a trapdoor function.



Symmetric Vs Asymmetric Key Cryptography



Symmetric	Asymmetric
Public Key Cryptography	Private Key Cryptography
Key exchange is a big problem.	No key exchange
Length of the key is small.	Compute-intensive because the length of the keys is usually large.
Encryption and decryption is faster.	Encryption and decryption is slower
Appropriate for long messages because speed of encryption/decryption is fast.	Appropriate for short messages because speed of encryption/decryption is slow.

Symmetric Vs Asymmetric Key Cryptography

Symmetric	Asymmetric
Symbols in plaintext and ciphertext are permuted and substituted.	Plaintext and ciphertext are treated as integers.
Used for encrypt/decrypt with asymmetric used to share and agree upon the key used.	Strongly used when parties involved have no prior relationship.
Number of keys required are more. $n*(n-1)/2$	Two keys per participant.



Symmetric Vs Asymmetric Key Cryptography

Table 2-5. *Key Requirements Comparison for Symmetric and Asymmetric Key Techniques*

Number of Participants	Number of Symmetric Keys	Number of Asymmetric Keys
2	1	4
4	6	8
10	45	20
50	1225	100
100	4950	200
1000	499500	2000

Game Theory

- Introduced by John Von Neumann.
- It is theory on games, where the games are not just what children play.
- Most situations where two or more parties are involved with some strategic behavior.
- A “game” can be defined as situations involving a “correlated rational choice”.
- It is a strategies involved in complex games with the art of making the best move, or opting for a best strategy in a given situation based on the objective.
- It may be the best move for the objective you have in mind which is usually finding an optimal solution.



Computer Science Engineering

- It is a clever engineering which stitches the components of cryptography, game theory and many others to build a blockchain.
- The computer science components that are used in blockchain are:
 - 1) Hash Pointers
 - 2) MerkleTrees



Hash Pointers

- Blockchain is a chain of blocks linked together.
- A block may consist of single or multiple transactions.
- A hash pointer is a cryptographic hash pointing to a data block, where the hash pointer is the hash of the data block.
- They point to the previous data block and provides a way for data verification.
- The purpose of the hash pointer is to build a tamper resistant blockchain that can be considered as a single source of truth.

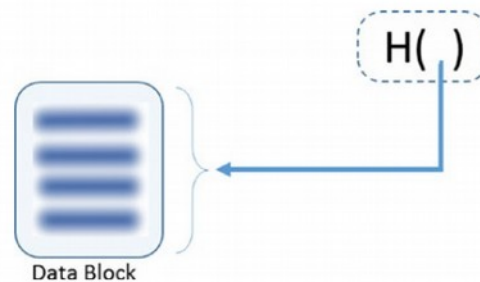


Figure 2-19. Hash pointer for a block of transactions

Hash Pointers

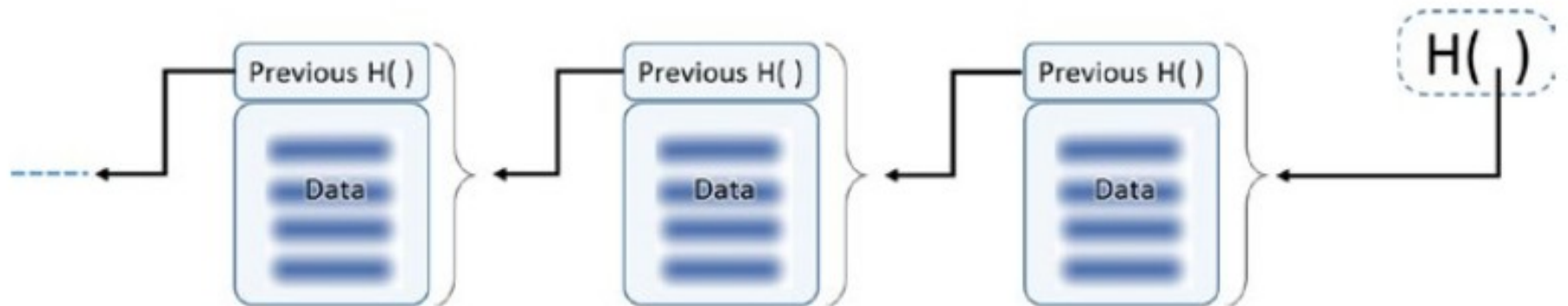
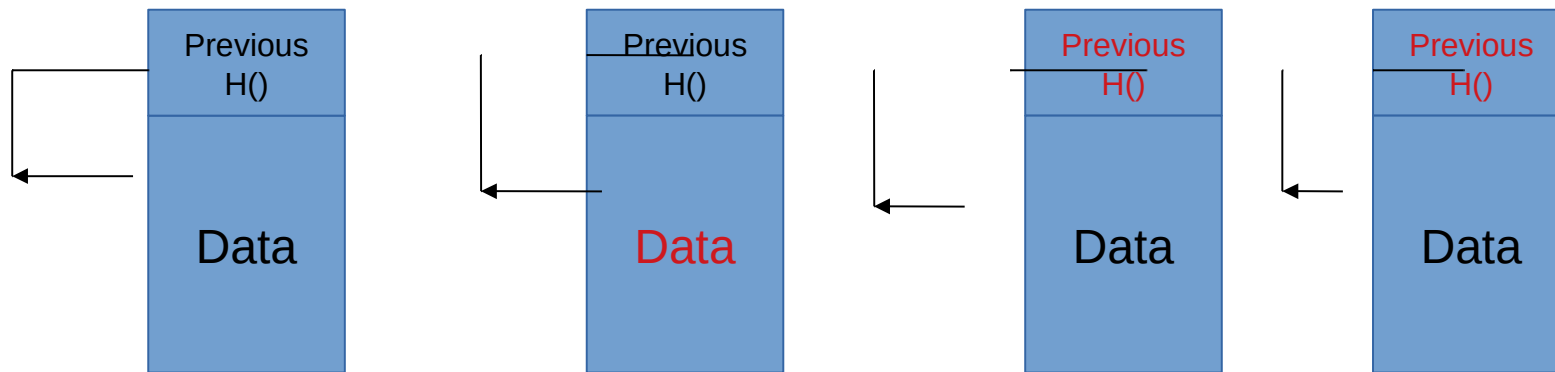


Figure 2-20. *Blocks in a blockchain linked through hash pointers*



Hash Pointers

- If someone tries to alter any transaction then there will be change in the hash of the block which is stored in the header of the next block.



Hash Pointers

- Each block is uniquely identified by its hash.
- To calculate hash, blockchain uses SHA-256 cryptography.
- The output hash produces a 256-bit hash output, represented as 64 characters, such as:
 - 0000000000000000a73b6a2af7bad40ec3fc2a83dafd76ef15f3d1b71a7132765



Merkle Trees

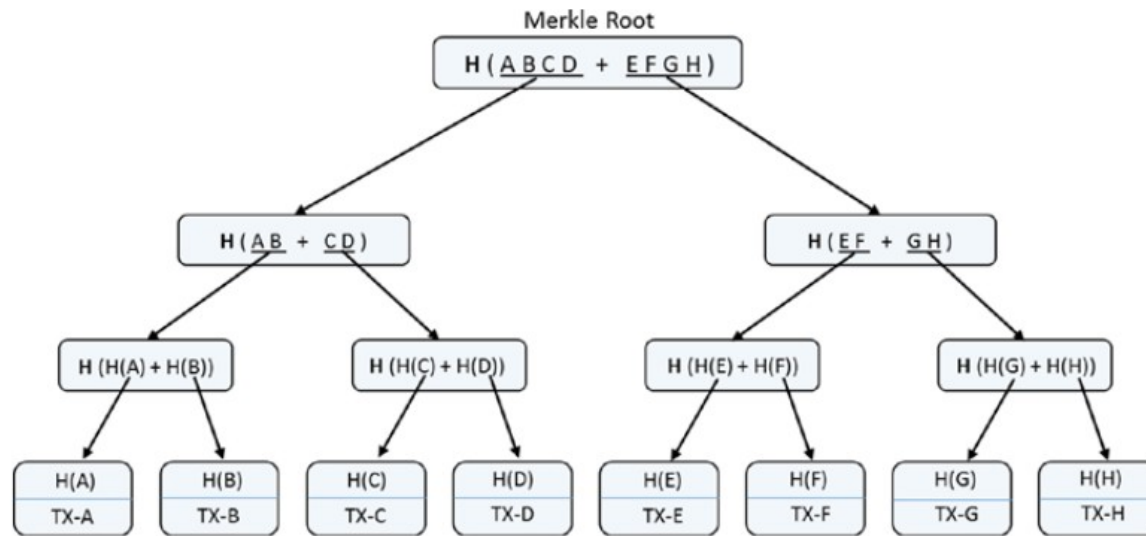


Figure 2-22. Merkle tree representation

Merkle Trees

- A Merkle tree is a binary tree of cryptographic hash pointer, hence it is a binary hash tree.
- It is named after its inventor Ralph Merkle.
- It is a data structure used in bitcoin.
- It is constructed by hashing paired data (usually transactions at the leaf level), then again hashing the hashed outputs all the way to root node called Merkle root.
- It is constructed in bottom-up manner.
- The leaves are always transactions of a single block in blockchain.
- It is tamper proof.
- It always has even number of items at the leaf level.

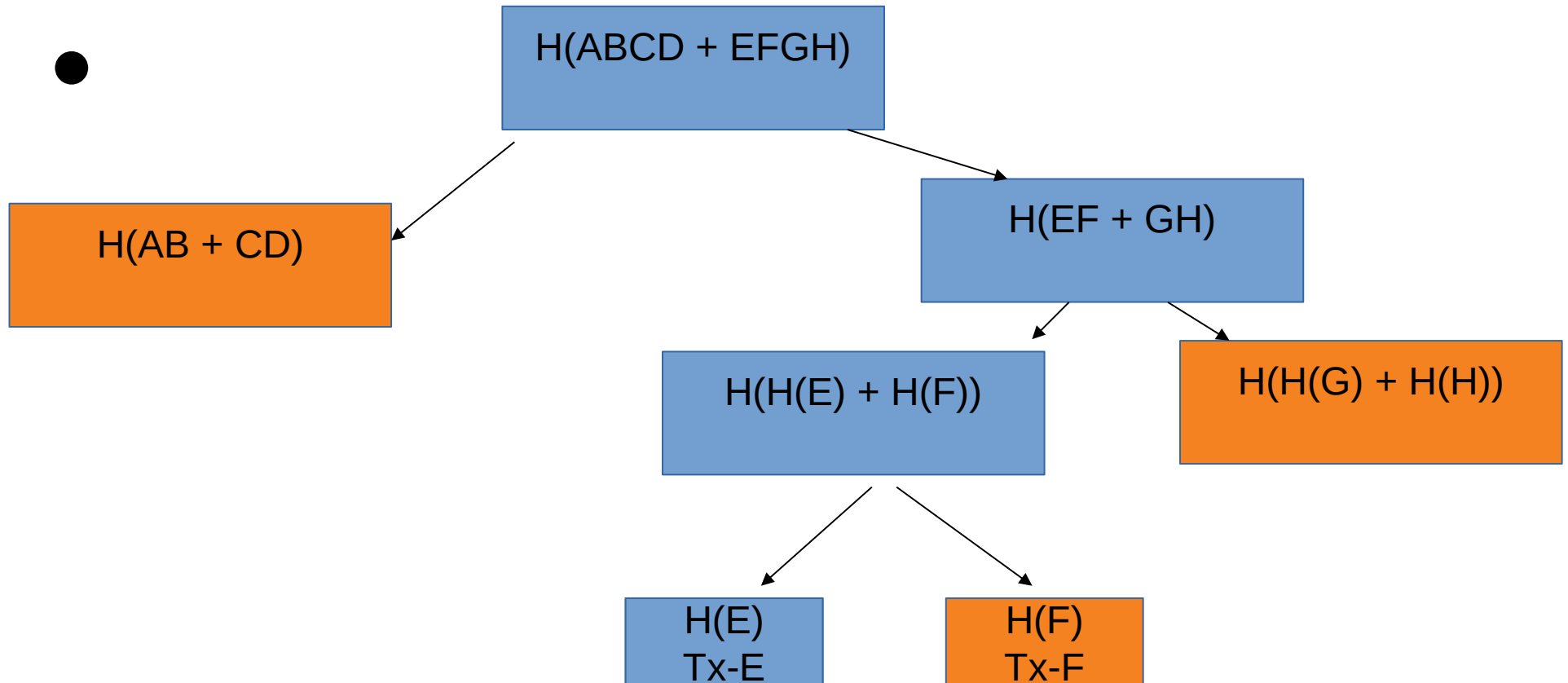


Merkle Trees

- A Merkle tree provides a very efficient way to verify if a specific transaction belongs to a particular block.
- If there are n transactions in a Merkle tree (leaf items), then this verification takes just $\text{Log}(n)$ times computations to verify the transaction, such as if $n = 8$, $\log_2 8 = 3$.
- $X = \log_2 n$
- $2^x = n$
- Two verifications need to be done: transaction as a part of the block and block as a part of the blockchain.
- The use of Merkle trees is not just limited to blockchains, they are used in many applications such as BitTorrent, Cassandra, Apache Wave



Merkle Trees



Putting it all together

- Cryptographic functions are one-way and cannot be inverted. They are deterministic and produce the same output for a given input.
- Using public key cryptography, digital signatures are possible. It helps in verifying the authenticity of the person/entity that has signed.
- The blockchain data structure, by using the cryptographic hashes, provides a tamper resistant chain of blocks. The usage of Merkle trees makes the transaction verification easier and faster.



Merkle Trees

- A Merkle tree provides a very efficient way to verify if a specific transaction belongs to a particular block.
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- Two verifications need to be done: transaction as a part of the block and block as a part of the blockchain.
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Merkle Trees

- A Merkle tree provides a very efficient way to verify if a specific transaction belongs to a particular block.
- If there are n transactions in a Merkle tree (leaf items), then this verification takes just $\log_2(n)$ times, such as if $n = 8$, $\log_2 8 = 3$.
- $X = \log_2 n$
- $2^x = n$
- Two verifications need to be done: transaction as a part of the block and block as a part of the blockchain.
- The use of Merkle trees is not just limited to blockchains, they are used in many applications such as BitTorrent, Cassandra, Apache Wave



Properties of Blockchain Solutions

● Some of the desired properties of blockchains:

- 1) Immutability
- 2) Forgery Resistant
- 3) Democratic
- 4) Double-Spend Resistant
- 5) Consistent State of the Ledger
- 6) Resilient
- 7) Auditable



Immutability

- Once transaction is recorded, it cannot be altered.
- Once the transaction is broadcast to the network, then almost everyone has a copy of it.
- As the chain of blocks grows bigger, it becomes completely immutable as it is practically not feasible to alter data in a block series.
- This leads to cryptographically secure.
- Any transaction that gets logged remains forever in the system.



Forgery Resistant

- A decentralized systems where transactions are public is more prone to different kinds of attacks.
- Cryptographic hash and digital signatures can be used to ensure the system is foregery resistant



Democratic

- Any peer-to-peer decentralized system should be democratic by design (except the private blockchains).
- There should not be any entity in the system that is more powerful than others.
- Every participant has equal rights, any decision made should be with consensus .



Double-Spend Resistant

- In a cryptocurrency settings, a double-spend attempt is when someone tries to spend the same amount to multiple people.
- Cryptography can ensure authenticity but does not helps to prevent double-spend .
- The only way possible to prevent it is to be aware of all the transactions
- So that the node that validates the transaction should definitely be accessible to the whole blockchain data till genesis block.



Consistent State of the Ledger

- The ledger is consistent throughout, to some extent.
- This is handled by using consensus algorithm.



Resilient

- The network should be resilient enough to withstand: -
 - 1) temporary node failures,
 - 2) unavailability of some computing nodes at times,
 - 3) network latency and
 - 4) packets drops.



Auditable

- A blockchain is a chain of blocks linked together through hashes.
- Its audibility already exists as the transactions blocks are linked till the genesis block.



Blockchain Transactions

- A transaction goes through series of steps to ultimately resides in the blockchain.



Distributed Consensus Mechanisms

- Grouping transactions into blocks is important for obvious reasons and consensus is required on a block by block basis.
- So, if any one node proposes a block and the rest of the nodes agree on it, then all those nodes add that block to their respective blockchains.
- Then who might be that lucky node to propose a block?
- This problem is solved by consensus mechanism.
- This can also be solved by reward/punishment basis where reward for honest behavior and punish for fraudulent activities.
- The goal of consensus is also to ensure that the network is robust enough to sustain various types of attacks.



Proof of Work

- The idea behind the PoW algorithm is that certain work is done for a block of transactions before it gets proposed to the whole network.
- A PoW is actually a piece of data that is difficult to produce in terms of computation and time, but easy to verify.
- One of the old usages of PoW was to prevent email spams.
- If a certain amount of work is to be done before one can send an email, then spamming a lot of people would require a lot of computation to be performed.
- Some amount of compute-intensive work is to be performed before producing a block, then it can help in two ways: one is that it will definitely take some time and the second is, if a node is trying to inject a fraudulent transaction in a block, then rejection of that block by the rest of the nodes will be very costly.
- The nodes which are able to solve a puzzle and propose a block are selected randomly.



Proof of Stake

- The Proof of Stake (PoS) algorithm is another consensus algorithm that is quite popular for distributed consensus.
- It isn't about mining, but is about validating blocks of transactions.
- There are no mining rewards due to generation of new coins, there are only transaction fees for the miners.
- In PoS systems, the validators have to bond their stake to be able to participate in validating the transactions.
- The probability of a validator producing a block is proportional to their stake; the more the amount at stake, the greater is their chance to validate a new block of transactions.
- A miner only needs to prove they own a certain percentage of all coins available at a certain time in a given currency system.
- There are variants of PoS algorithm such as naive PoS, delegated PoS, chain-based PoS, BFT-style PoS, and Casper PoS.
- Delegated PoS (DPOS) is used by Bitshares and Casper PoS is being developed to be used in Ethereum..
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Blockchain Applications

- There are applications being built that treat blockchain as a backend database behind a web server, and there are applications that are completely decentralized with no centralized server.
- Bitcoin blockchain, for example, is a blockchain application where there is no server to send a request to.
- It is a public blockchain.
- Usually for such public blockchains, usage of resources from cloud service providers such as Microsoft Azure, IBM Bluemix, etc. are not quite popular yet.
- Applications built on private blockchain which are hosted by some cloud service providers by consuming Blockchain-as-a-Services (BaaS).
- Businesses are using blockchain solutions in the cloud with “pay as you use” model to achieve transparency among them.



Blockchain Applications

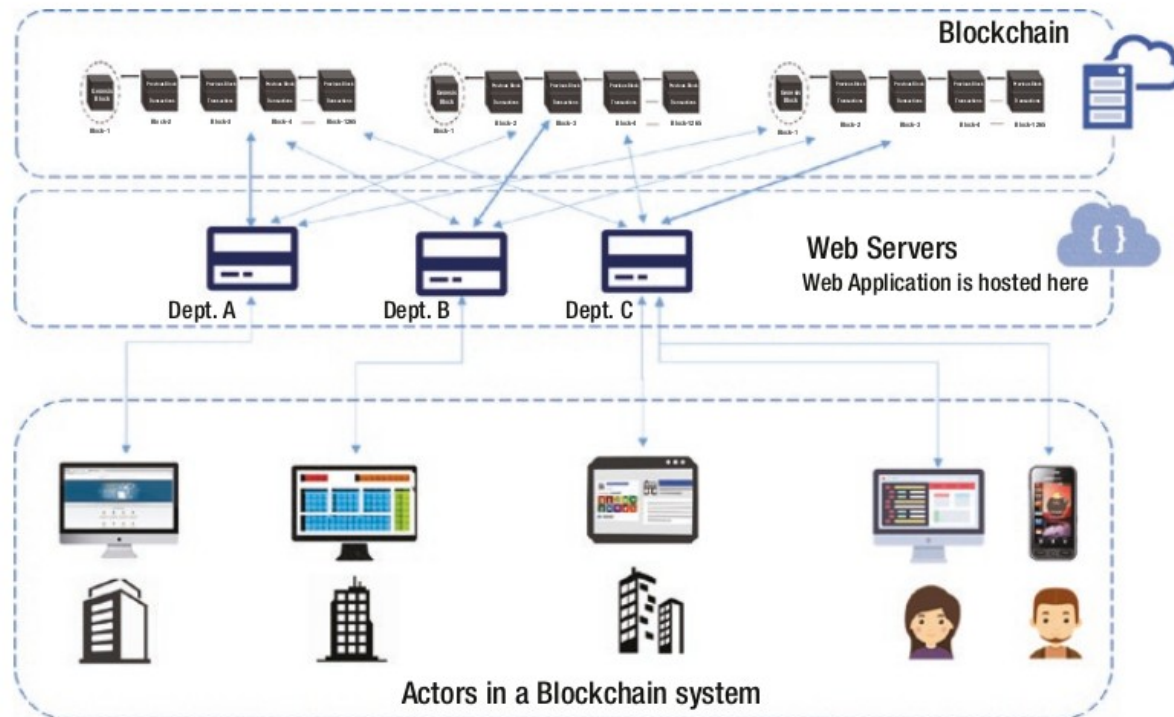


Figure 2-26. *Cloud-powered blockchain system*

Blockchain Applications

- There are also decentralized applications (DApps) being built on Ethereum blockchain networks. These applications could be permissioned on private Ethereum or could be permissionless on a public Ethereum network.
- Pure blockchain native applications could be built.
- Applications that treat blockchain as just a backend are also being built, and there are hybrid applications that are also being built that use the legacy applications and use blockchain for some specific purpose only.



Blockchain Applications

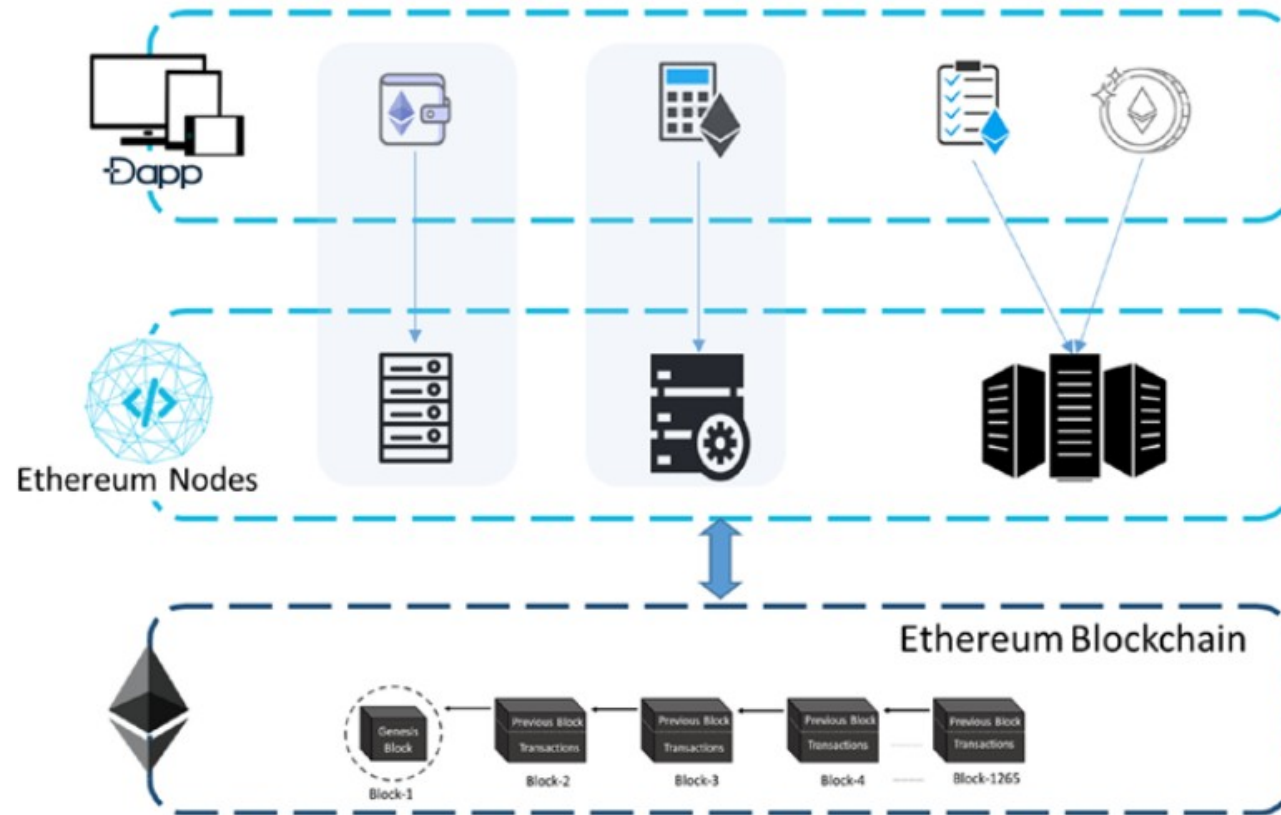


Figure 2-27. *DApps on Ethereum network*